

EduGenie-Google Gemini Powered Learning Assistant

Government Arts and Science College-Manur-MSU142

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Project Planning Phase

- **Phase Overview:** Translates project initiation approval into a structured, executable roadmap before active development begins.
- **Primary Purpose:** Establishes formal baselines for scope, time, cost, and quality to measure progress and control variances.
- **Core Objective:** Identifies technical, financial, and operational risks early to implement proactive mitigation strategies.

Scope Management & Work Breakdown Structure (WBS)

- **Scope Definition:** Sets exact project boundaries by defining deliverables, acceptance criteria, and explicit out-of-scope items.
- **Scope Creep Prevention:** Establishes a formal change control process to evaluate and approve any proposed scope adjustments.
- **WBS Decomposition:** Breaks high-level project objectives down into smaller, actionable work packages (Phase → Deliverable → Work Package).
- **Work Package Assignment:** Ensures every single leaf-level WBS element has a clear deliverable and assigned owner.

Schedule Management & Task Sequencing

- **Activity Identification:** Lists all specific tasks required to complete every work package defined in the WBS.
- **Dependency Mapping:** Defines technical sequence relationships between activities (e.g., Finish-to-Start, Start-to-Start).
- **Duration Estimation:** Calculates effort hours and calendar duration for each task using expert judgment or historical data.
- **Critical Path Method (CPM):** Identifies the longest sequence of dependent tasks that dictates the minimum overall project timeline.
- **Milestone Scheduling:** Sets fixed calendar targets for major progress checkpoints, reviews, and client sign-offs.

Resource Allocation & Budget Baseline

- **Resource Assignment:** Maps human skill sets, software tools, hardware, and facilities directly to scheduled tasks.
- **RACI Matrix Setup:** Clarifies team responsibilities by designating who is **R**esponsible, **A**ccountable, **C**onsulted, and **I**nformed.
- **Cost Estimation:** Aggregates direct labor rates, equipment purchases, licensing fees, and operational overhead.

- **Contingency Buffer:** Reserves a financial buffer (typically 10–15%) for unexpected technical or operational costs.
- **Cost Baseline Approval:** Locks the approved financial allocation to track expenditure against actual project progress.

Risk Assessment & Quality Control

- **Risk Identification:** Brainstorms potential technical bottlenecks, vendor delays, resource turnover, and budget overruns.
- **Probability & Impact Scoring:** Evaluates each risk on occurrence likelihood and potential severity to prioritize focus.
- **Mitigation Planning:** Assigns specific preventive actions, trigger conditions, and fallback plans in the Risk Register.
- **Quality Assurance Rules:** Outlines inspection standards, code/design review gates, and formal acceptance criteria.
- **Communication Matrix:** Defines meeting cadences, weekly reporting formats, and stakeholder escalation paths.